

SP ENGINEERING ACADEMY

Video Lectures / Course Materials

COMPLETE COURSE

DIGITAL SYSTEMS

DIGITAL CIRCUIT DESIGN

Part – VI – Logic Gates and its Implementation

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1. Digital Logic Gates

Video Link - <https://youtu.be/q6QT5RNu8UI>

Basic Gates - AND gate, OR gate and NOT gate

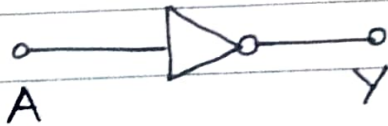
The three primitive logic functions are represented by distinctive logic symbols known as basic gates

Derived Logic Gates - NAND gate, NOR gate, EX-OR gate
EX-NOR gate

From these primitive logic functions we have derived logic functions and hence derived logic gates.

(a) NOT gate (Inverter)

The output is a complement of input



$$Y = \bar{A}$$

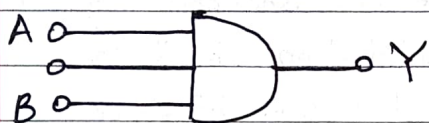
Input-	Output
A	Y
0	1
1	0

Application

Used to Complement-(inverter) digital signal

(b) AND gate

The output is high only when all inputs are high



$$Y = A \cdot B$$

Input		Output
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Application : Used to implement logical AND operation

(c) OR gate

The output is high when any of the input is high



$$Y = A + B$$

Application:

Used to implement logical OR operation

Input		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(d) Buffer

The output is same as input



$$Y = A$$

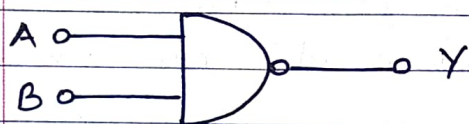
Input	Output
A	Y
0	0
1	1

Application:

It is used to increase output driving Capacity

(e) NAND gate

Video Link - <https://youtu.be/0qcAcLKb1bs>



$$Y = \overline{A \cdot B}$$

Input		Output
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

The output is high only when one of the input is low

Application: It can be used to implement any digital circuit

(f) NOR gate

The output is high when all the inputs are low



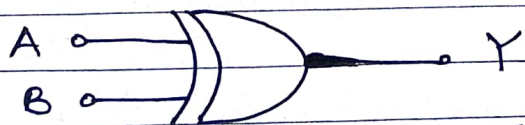
$$Y = \overline{A + B}$$

Input		Output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Application: It can be used to implement any digital circuit

(g) Exclusive OR (EX-OR) gate

The output is high only when odd number of inputs are high



$$Y = A \oplus B = \bar{A}B + A\bar{B}$$

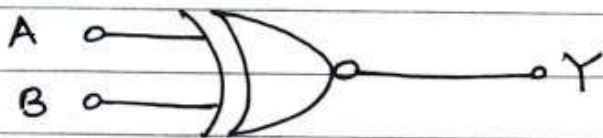
Input		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Application:

It is used to implement magnitude comparator, gray code, Converter, adder/subtractor circuits, parity generator, modulo-2 adder, etc.

(R) Exclusive NOR (EX-NOR) gate

The output is high only when even number of ones at the input or all inputs are high



$$Y = \overline{A \oplus B}$$
$$= \bar{A} \cdot \bar{B} + A \cdot B$$

Input		Output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Application:

It is used to implement even parity generator, Comparator, even parity checker etc.

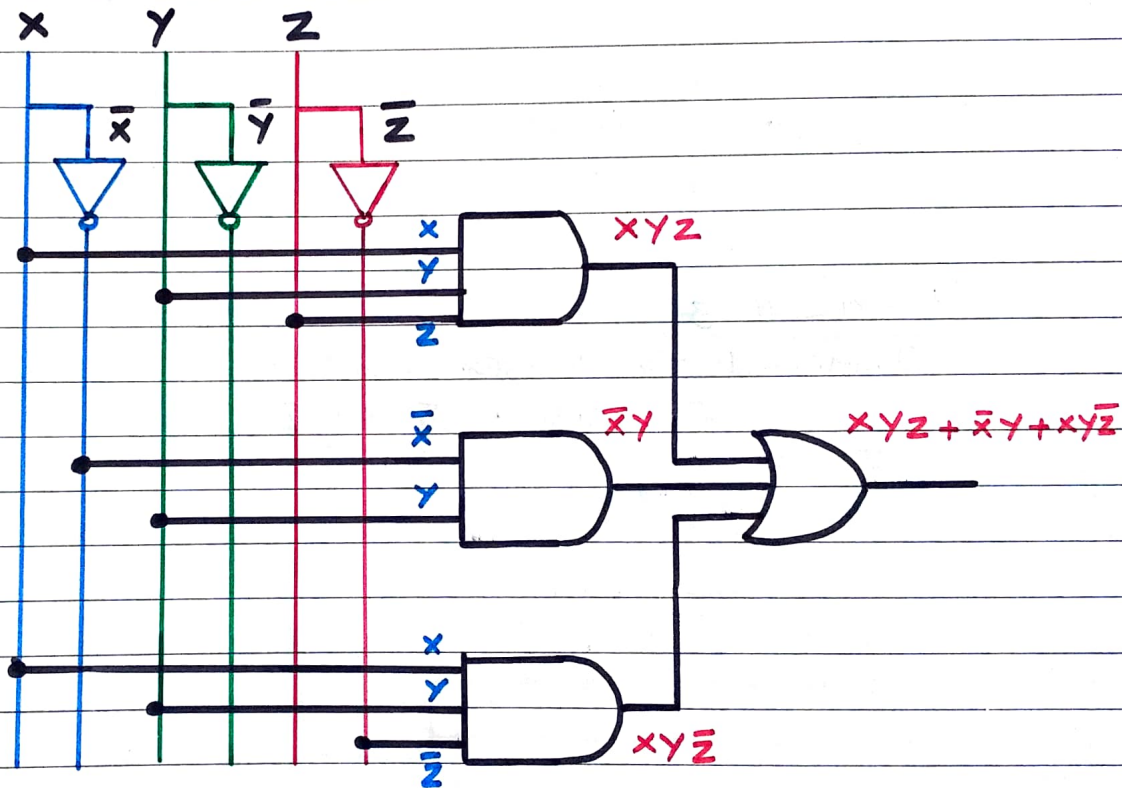
Realizing Logic Function with Gates

The boolean algebra is used to express the output of any Combinational network. Such a network can be implemented using logic gates.

Example # 01

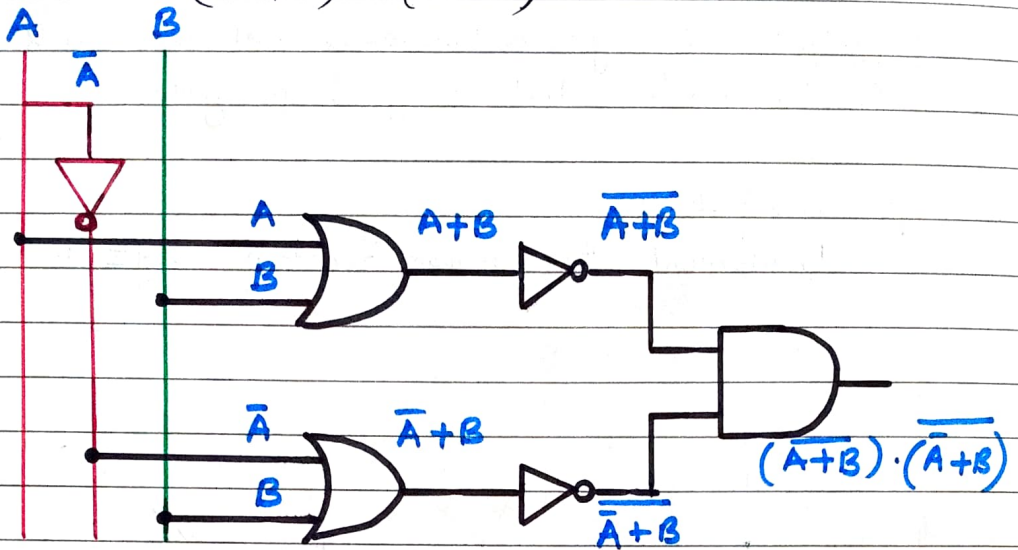
Video Link - <https://youtu.be/a2e556zSFYE>

Implement the Boolean function $xyz + \bar{x}y + x\bar{y}\bar{z}$



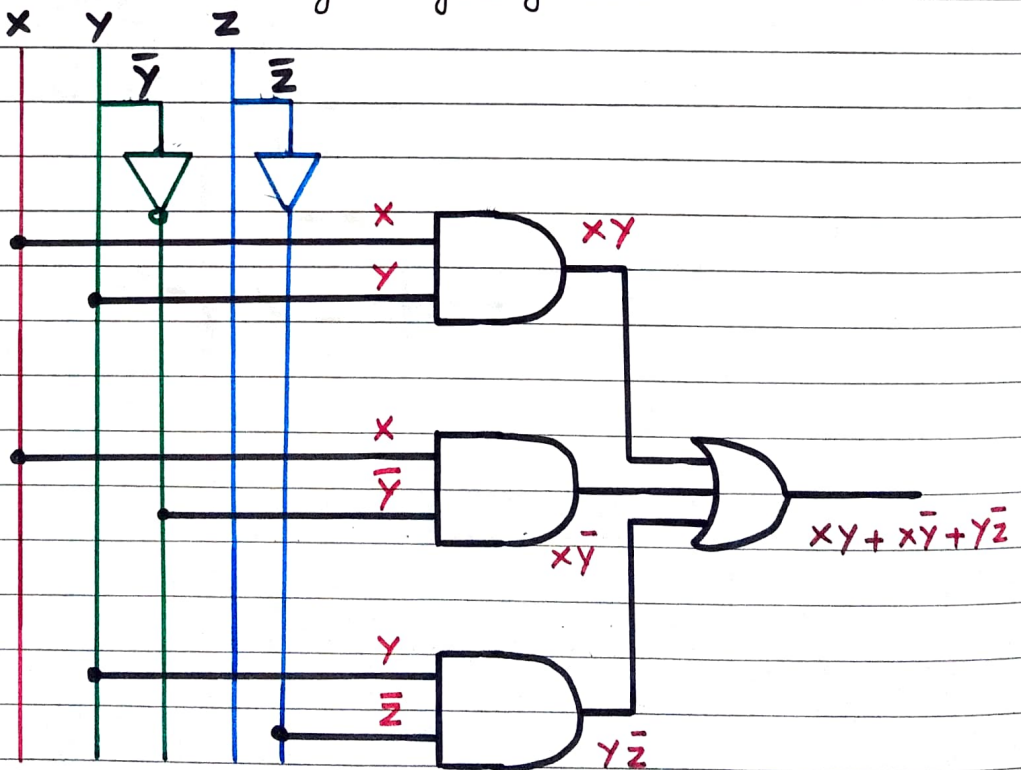
Example #02

Implement the Boolean Function
 $(\overline{A+B}) \cdot (\overline{A \cdot B})$



Example #03

Implement the Boolean Function
 $F = xy + x\overline{y} + y\overline{z}$



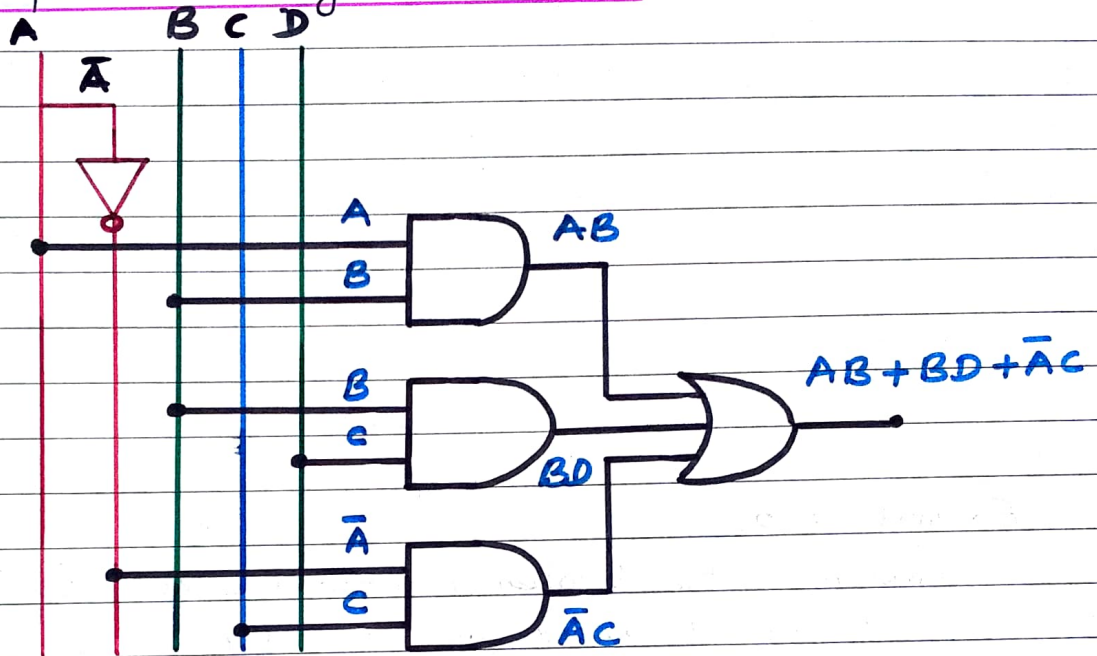
Example # 04

Video Link - <https://youtu.be/dP-hJiZD4a0>

For the following function implement the SOP and POS Circuit

$$F = AB + BD + \bar{A}C$$

Implement using SOP Circuit



Implement using POS Circuit

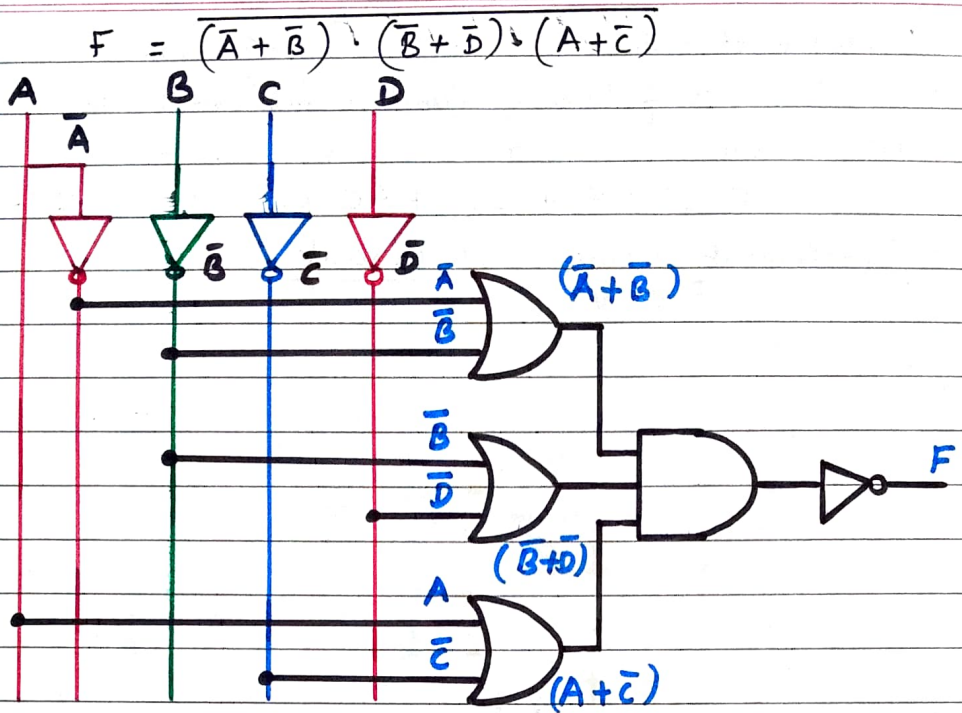
$$F = \overline{\overline{F}}$$

$$= \overline{\overline{AB + BD + \bar{A}C}}$$

$$= \overline{\bar{A}\bar{B} \cdot \bar{B}\bar{D} \cdot \bar{\bar{A}}\bar{C}}$$

$$= \overline{(\bar{A} + \bar{B}) \cdot (\bar{B} + \bar{D}) \cdot (\bar{\bar{A}} + \bar{C})}$$

$$= \overline{(\bar{A} + \bar{B}) \cdot (\bar{B} + \bar{D}) \cdot (A + \bar{C})}$$



Example #05

Video Link - <https://youtu.be/Gj9gvliVLcA>

Implement the Boolean Function $F = xy + \overline{x}y + \overline{y}z$

- (i) Implement with only OR and NOT gate
- (ii) Implement with only AND and NOT gate

$$F = \overline{\overline{F}}$$

$$= \overline{xy + \overline{x}y + \overline{y}z}$$

$$= \overline{(xy) \cdot (\overline{\overline{xy}}) \cdot (\overline{\overline{\overline{y}z}})}$$

$$= \overline{(xy) \cdot (\overline{\overline{xy}}) \cdot (\overline{\overline{\overline{y}z}})}$$

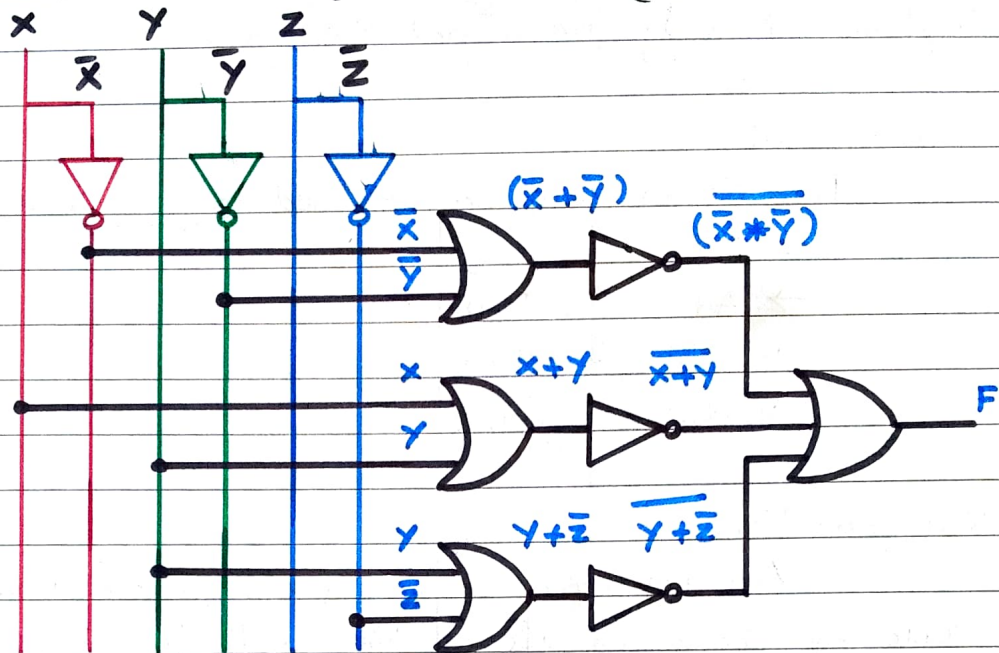
$$= \overline{(\overline{x} + \overline{y}) \cdot (\overline{\overline{x} + \overline{y}}) \cdot (\overline{\overline{\overline{y} + \overline{z}}})}$$

$$= \overline{(\overline{x} + \overline{y}) \cdot (x + y) \cdot (y + \overline{z})}$$

$$F = \overline{(\overline{x} + \overline{y})} + \overline{(x + y)} + \overline{(y + \overline{z})}$$

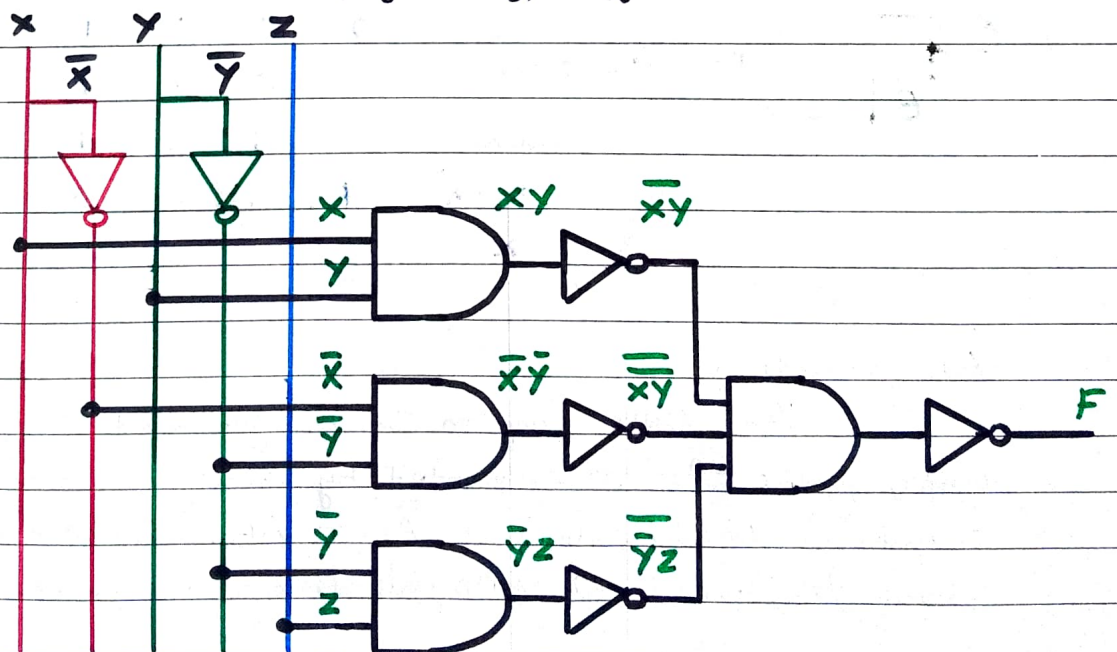
(i) Implement with only OR and NOT Gate

$$F = (\bar{x} + \bar{y}) + (x + y) + (xy + \bar{z})$$



(ii) Implement with only AND and NOT Gate

$$F = (\bar{x}y) \cdot (\bar{x}\bar{y}) \cdot (\bar{y}z)$$



Universal Gate

The NAND and NOR gates are known as Universal gates, since any logic function can be implemented using NAND and NOR gates.

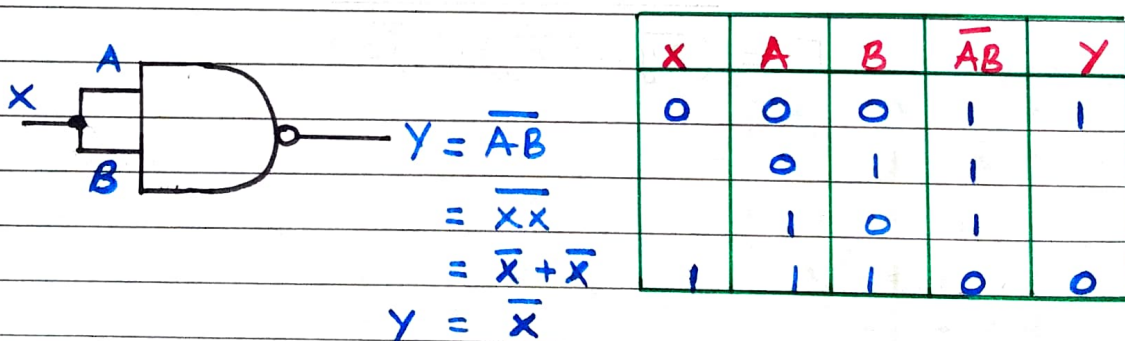
NAND Gate Video Link - <https://youtu.be/vsffF9ZGJ00>

The NAND gate can be used to generate the

- NOT function
- AND function
- OR function
- NOR function

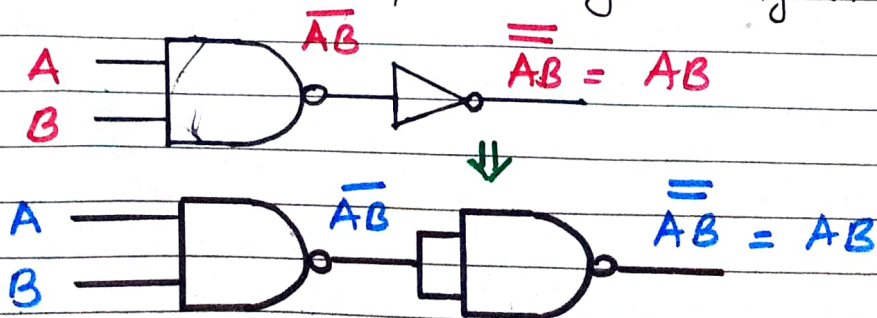
NOT Function

An inverter can be made from a NAND gate by connecting all of the inputs together and creating, in effect, a single common input as shown below for a two input gate.



AND Function

An AND function can be generated using only NAND gates. It is generated by simply inverting the output of a NAND gate (i.e.) $\overline{\overline{AB}} = AB$. Diagram shows the two input AND gate using NAND gates.



A	B	AB
0	0	0
0	1	0
1	0	0
1	1	1

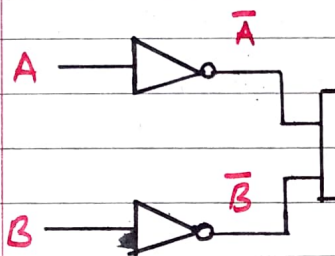
Truth Table

A	B	$\overline{A}B$	$A\overline{B}$
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

Truth Table

AND Gate Using NAND

OR Function





$$Y = \overline{\overline{A} \cdot \overline{B}}$$

$$= \overline{\overline{A}} + \overline{\overline{B}}$$

$$= A + B$$

A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

NOR Function

A	B	$\bar{A}$	$\bar{B}$	$\bar{A} \cdot \bar{B}$	$\bar{A} \cdot B$
0	0	1	1	1	0
0	1	1	0	0	1
1	0	0	1	0	1
1	1	0	0	0	1

$$\overline{A} \quad \overline{A} \cdot \overline{B} \quad \overline{A} \cdot \overline{B}$$

NOR Function

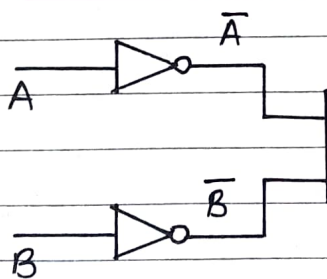


Diagram illustrating the implementation of De Morgan's Law for the expression $\overline{A \cdot B}$.

The circuit consists of two AND gates and one OR gate.

Input A is connected to the first AND gate. Input B is connected to the second AND gate.

The output of the first AND gate is $\overline{A}$.

The output of the second AND gate is $\overline{B}$.

The outputs $\overline{A}$ and $\overline{B}$ are connected to a third AND gate.

The output of the third AND gate is $\overline{\overline{A} \cdot \overline{B}}$.

The output of the third AND gate is connected to an OR gate.

The output of the OR gate is $\overline{A + B}$.

The final output is $\overline{A + B}$.

Handwritten notes show the simplification of the expression:

$$\overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A + B}} = A + B$$

A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

Truth Table

A	B	$\bar{A}$	$\bar{B}$	$\overline{A \cdot B}$	$\overline{\bar{A} \cdot \bar{B}}$
0	0	1	1	0	1
0	1	1	0	1	0
1	0	0	1	1	0
1	1	0	0	1	0

Truth Table

NOR Gate Using NAND

NOR Gate

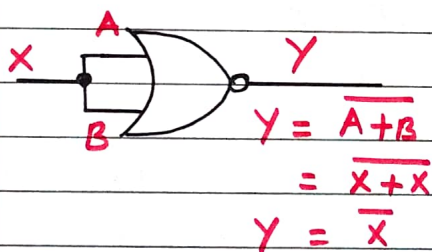
Video Link - <https://youtu.be/56Zl8VWD9l8>

The NOR gate is also a universal gate, since it can be used to generate

- NOT function
- OR function
- AND function
- NAND function

NOT Function

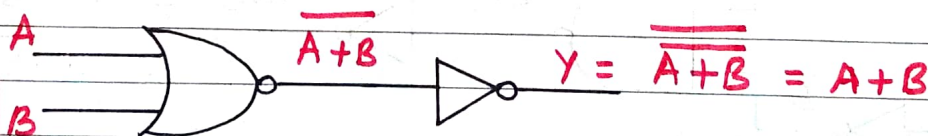
An inverter can be made from NOR gate by connecting all of the inputs together and creating, in effect, a single common input



X	A	B	$\overline{A+B}$	Y
0	0	0	1	1
	0	1	0	
	1	0	0	
1	1	1	0	0

OR Function

An OR function can be generated using only NOR gates. It can be generated by simply inverting output of NOR gate



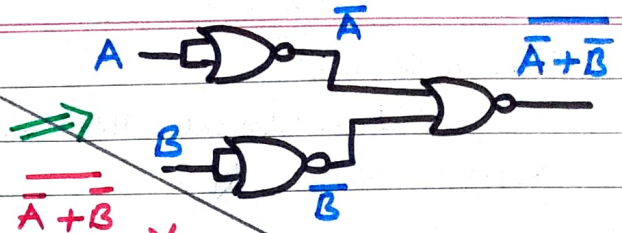
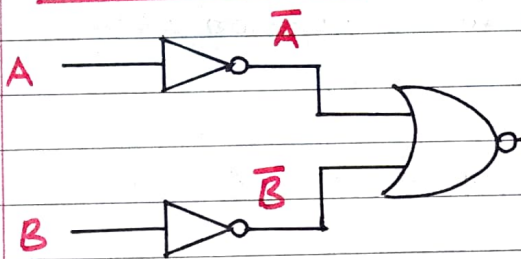
A	B	$A+B$
0	0	0
0	1	1
1	0	1
1	1	1



A	B	$\overline{A}$	$\overline{B}$	$\overline{A+B}$	$\overline{\overline{A+B}}$
0	0	1	1	1	0
0	1	1	0	0	1
1	0	0	1	0	1
1	1	0	0	0	1



AND Function

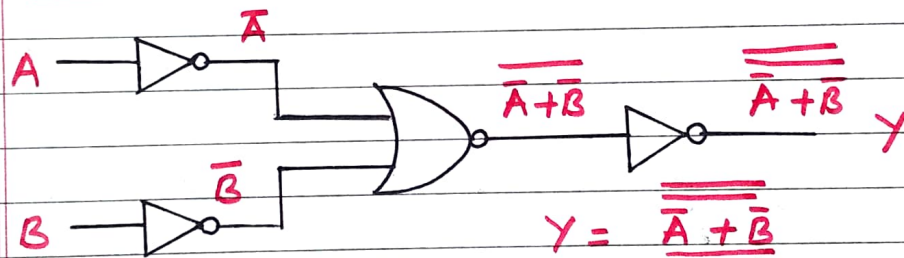


$$Y = \overline{\bar{A} + \bar{B}} = \bar{\bar{A}} \cdot \bar{\bar{B}} = A \cdot B$$

A	B	AB
0	0	0
0	1	0
1	0	0
1	1	1

A	B	$\bar{A} + \bar{B}$	$\overline{\bar{A} + \bar{B}}$
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

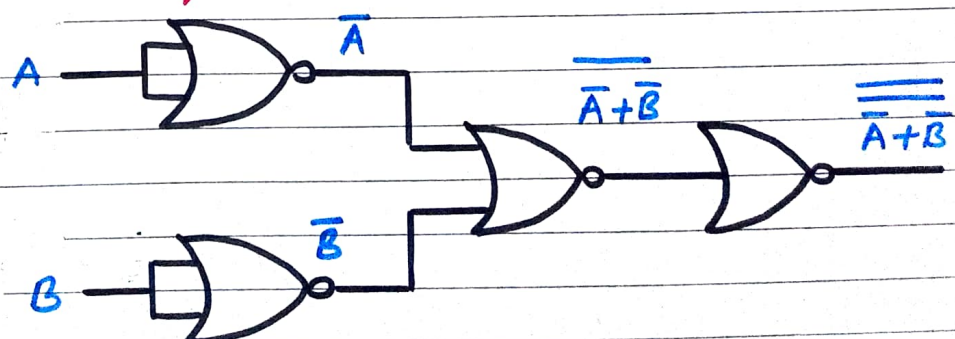
NAND Function



$$Y = \overline{\bar{A} + \bar{B}} = \bar{\bar{A}} \cdot \bar{\bar{B}} = A \cdot B$$

A	B	$\bar{A} \bar{B}$
0	0	1
0	1	1
1	0	1
1	1	0

A	B	$\bar{A} \bar{B}$	$\bar{A} + \bar{B}$	$\overline{\bar{A} + \bar{B}}$	$\overline{\overline{\bar{A} + \bar{B}}}$
0	0	1	1	0	1
0	1	1	1	0	1
1	0	1	1	0	1
1	1	0	0	1	0



Example 01

Video Link - https://youtu.be/G_dVBgHakws

Implement the following function with
NAND - NAND Logic

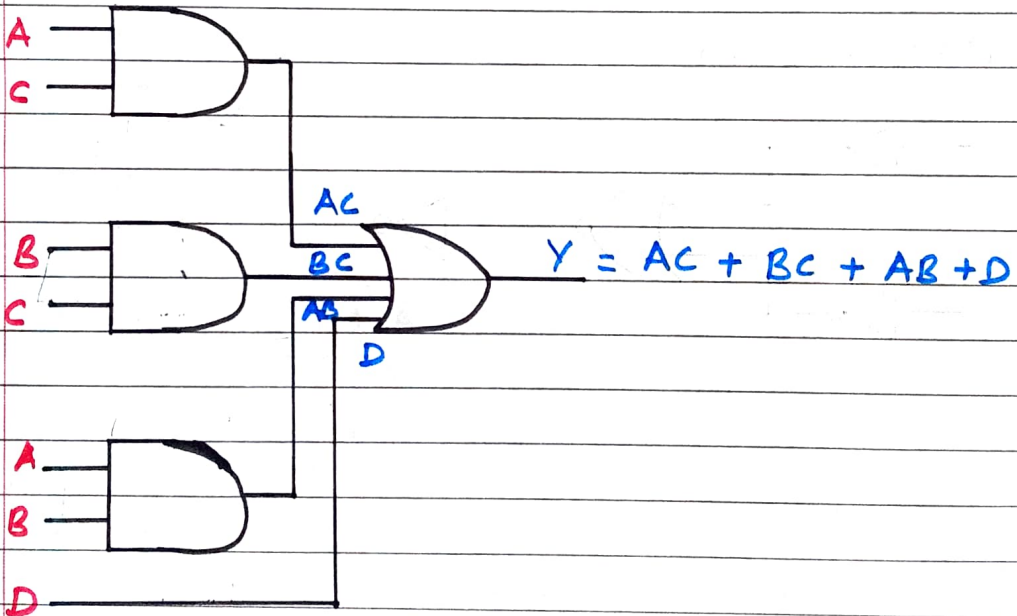
$$Y = AC + ABC + \bar{A}Bc + AB + D$$

$$Y = AC + ABC + \bar{A}Bc + AB + D$$

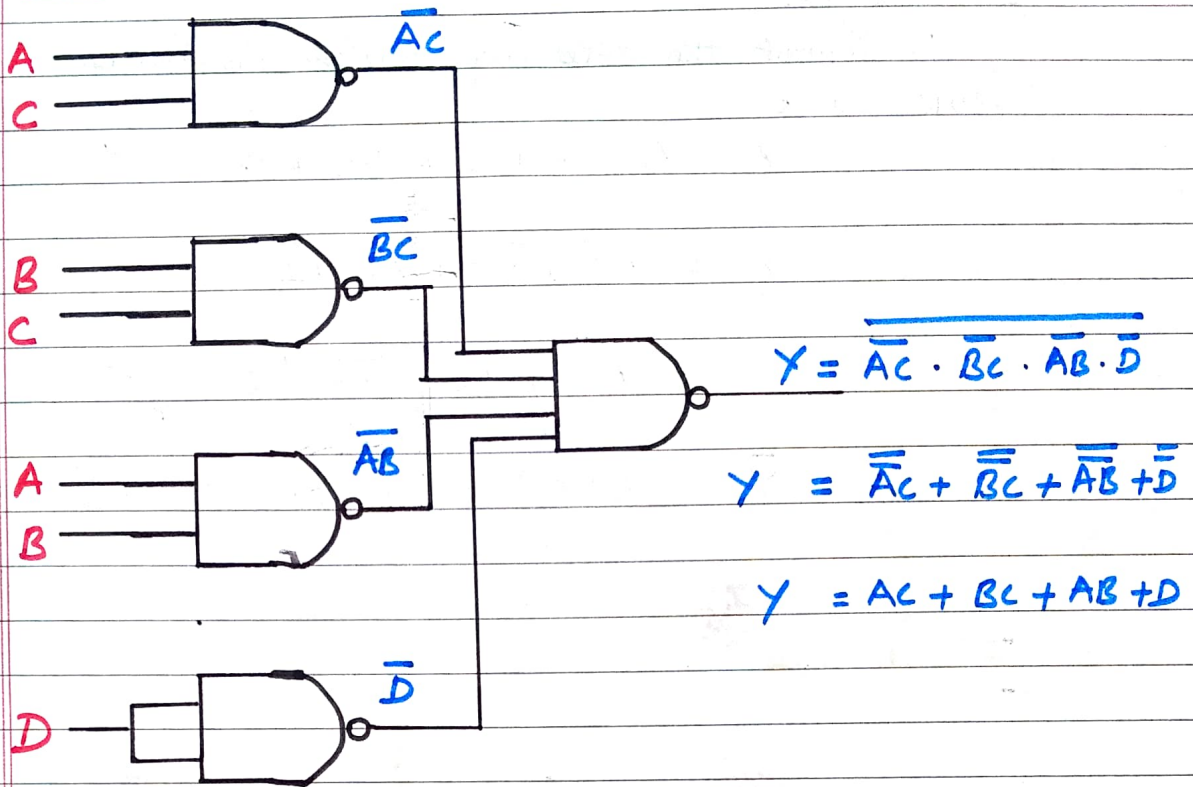
$$Y = AC + Bc(A + \bar{A}) + AB + D$$

$$Y = AC + Bc + AB + D$$

Implementation Using AND-OR Logic



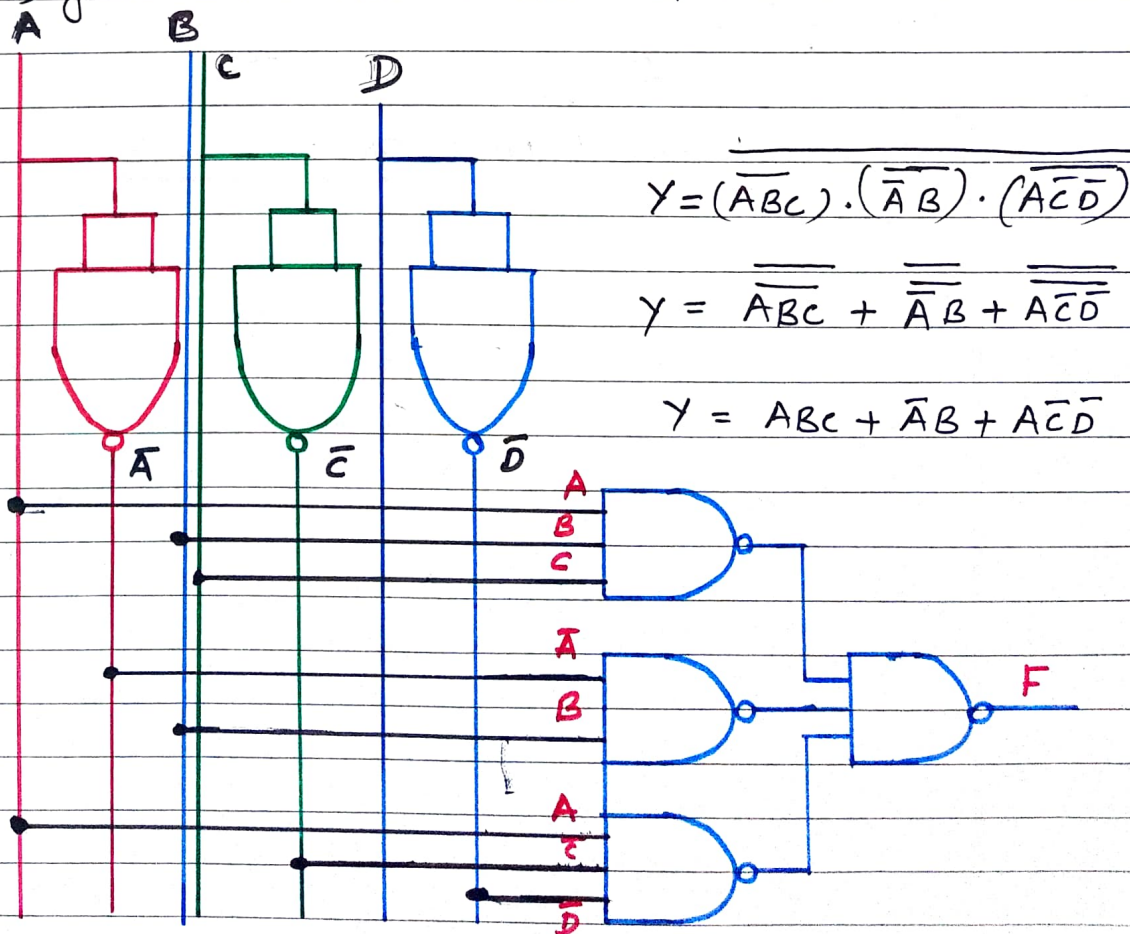
Implementation using NAND-NAND Logic



Example 02

Video Link - <https://youtu.be/upmWvQXtdTY>

Draw the logic circuit for following function using only NAND gates: $F = ABC + \overline{A}B + A\overline{C}\overline{D}$



Example 03Video Link - <https://youtu.be/GbTRYa-7xXs>

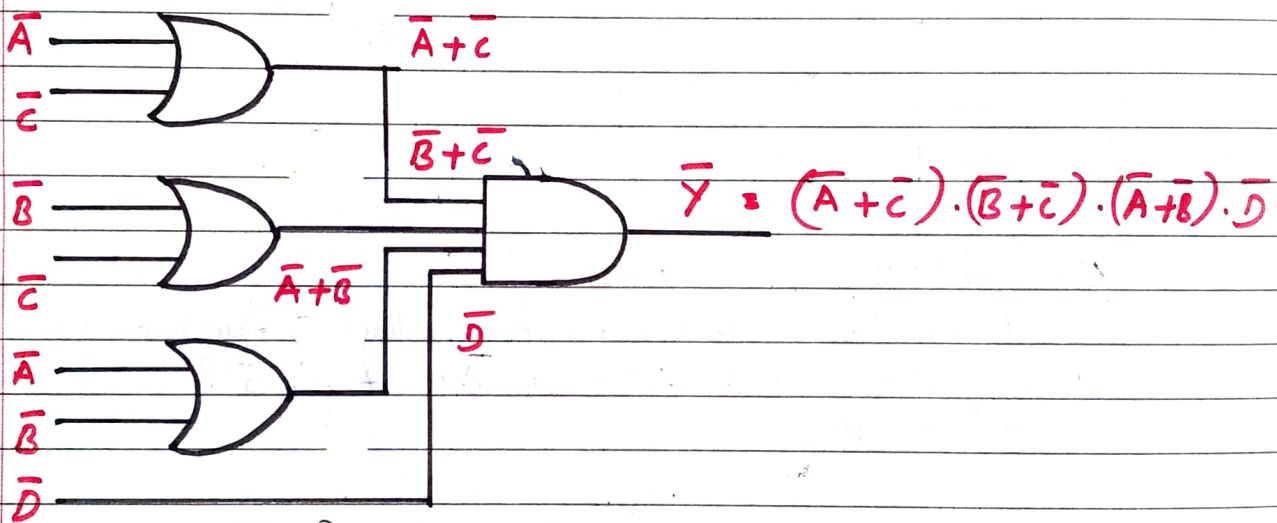
Implement the following Boolean function with NOR logic

$$Y = AC + BC + AB + D$$

$$\bar{Y} = \overline{AC + BC + AB + D}$$

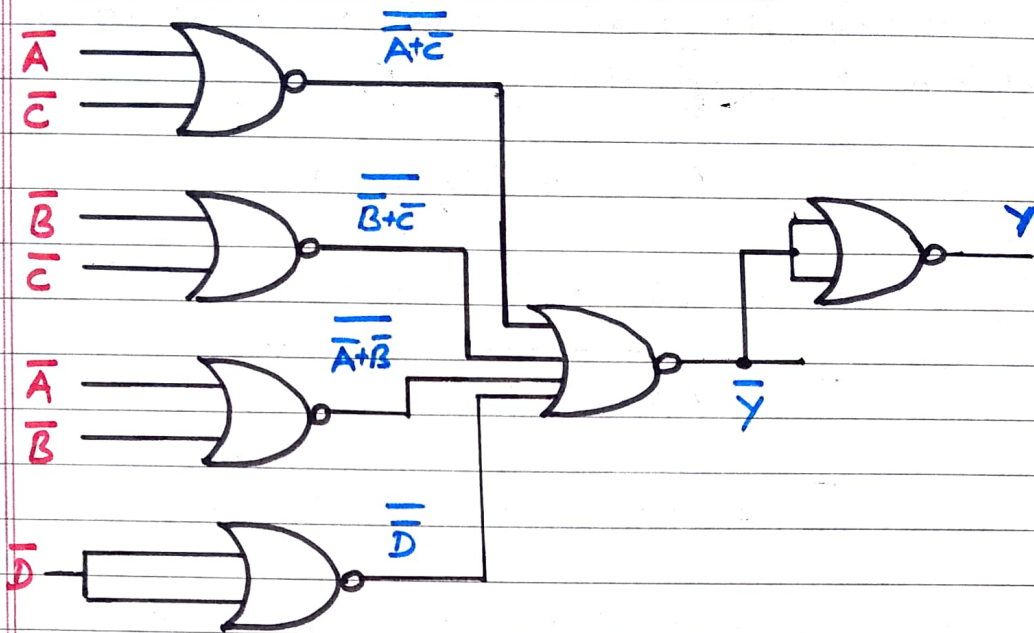
$$= (\bar{A}\bar{C}) \cdot (\bar{B}\bar{C}) \cdot (\bar{A}\bar{B}) \cdot \bar{D}$$

$$\bar{Y} = (\bar{A} + \bar{C}) \cdot (\bar{B} + \bar{C}) \cdot (\bar{A} + \bar{B}) \cdot \bar{D}$$



Implement with OR-AND Logic

Implement with NOR-NOR Logic



$$\bar{Y} = \overline{(\bar{A}\bar{C}) \cdot (\bar{B}\bar{C}) \cdot (\bar{A}\bar{B}) \cdot \bar{D}}$$

$$= \overline{(\bar{A} + \bar{C}) \cdot (\bar{B} + \bar{C}) \cdot (\bar{A} + \bar{B}) \cdot \bar{D}}$$

$$\bar{Y} = (A + C) \cdot (B + C) \cdot (A + B) \cdot D$$

$$\therefore Y = (A + C) \cdot (B + C) \cdot (A + B) \cdot D$$

$$\bar{Y} = \overline{\bar{A} + \bar{C} + \bar{B} + \bar{C} + \bar{A} + \bar{B} + \bar{D}}$$

$$= \overline{\bar{A}\bar{C} + \bar{B}\bar{C} + \bar{A}\bar{B} + \bar{D}}$$

$$\bar{Y} = AC + BC + AB + D$$

$$\therefore Y = AC + BC + AB + D$$

Example 04

Video Link - <https://youtu.be/XSnJM7kza5s>

Draw the logic circuit for following function using only NOR gates

$$F = AB\bar{C} + AB(C+D)$$

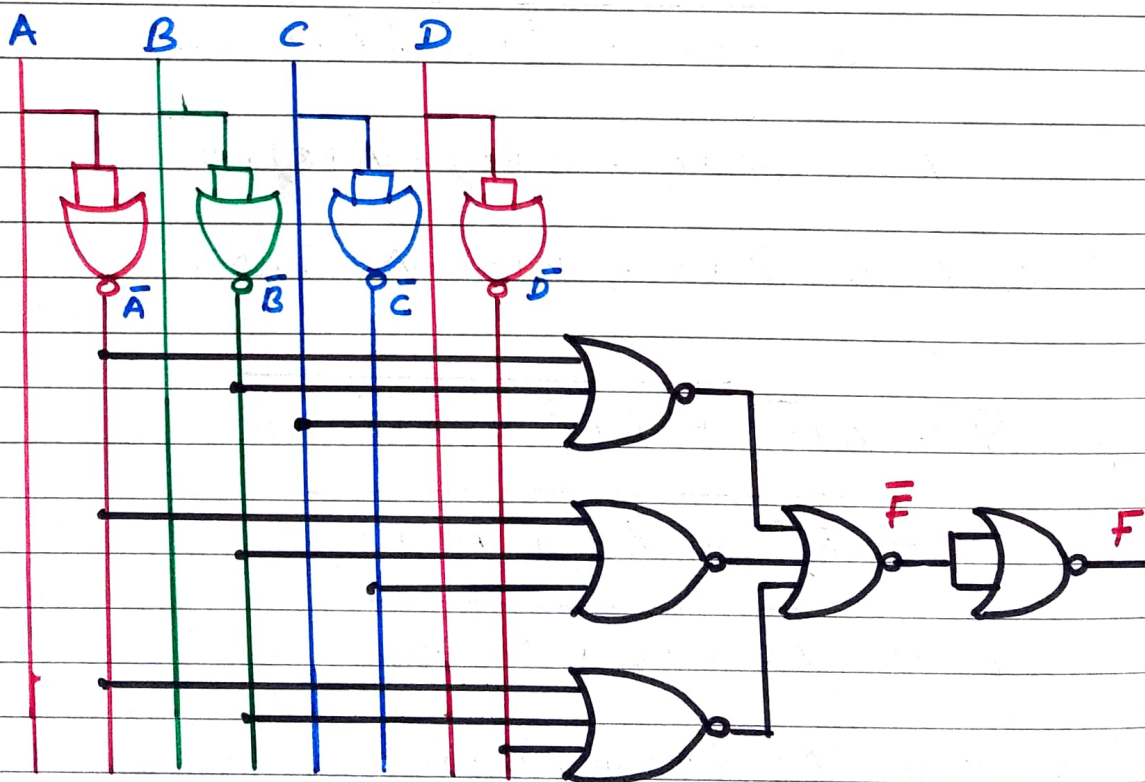
$$F = AB\bar{C} + ABC + ABD$$

$$\bar{F} = AB\bar{C} + ABC + ABD$$

$$\bar{F} = (\overline{AB\bar{C}}) \cdot (\overline{ABC}) \cdot (\overline{ABD})$$

$$F = (\bar{A} + \bar{B} + \bar{C}) \cdot (\bar{A} + \bar{B} + \bar{C}) \cdot (\bar{A} + \bar{B} + \bar{D})$$

$$\bar{F} = (\bar{A} + \bar{B} + C) \cdot (\bar{A} + \bar{B} + \bar{C}) \cdot (\bar{A} + \bar{B} + \bar{D})$$



$$\overline{F} = \overline{(\overline{A} + \overline{B} + c)} + \overline{(\overline{A} + \overline{B} + \overline{c})} + \overline{(\overline{A} + \overline{B} + \overline{D})}$$

$$\overline{F} = \overline{\overline{A} \overline{B} \overline{c}} + \overline{\overline{A} \overline{B} \overline{c}} + \overline{\overline{A} \overline{B} \overline{D}}$$

$$\overline{F} = (\overline{A} \overline{B} \overline{c}) (\overline{A} \overline{B} \overline{c}) (\overline{A} \overline{B} \overline{D})$$

$$\overline{F} = (\overline{A} + \overline{B} + c) (\overline{A} + \overline{B} + \overline{c}) (\overline{A} + \overline{B} + \overline{D})$$

$$\overline{F} = \overline{AB\overline{c}} + \overline{ABc} + \overline{ABD}$$

$$\therefore F = AB\overline{c} + ABc + ABD$$

$$F = AB\overline{c} + AB(c + D)$$